NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

# R23.4a Forecast uses (high-wind background fields, wind power, CFD)

**Group** AI Weather Models

**Status** Closed (negative result)

**Period** 2026-10-13 11:30 — 2026-10-21 09:56

## Stage conclusion

## Project report

# Uses of AI weather foundation models for nowcasting / short-range forecasting

Stop category: question answered (status: On hold · closure recommended, awaiting PI review; to be decided with a related PI decision item)

## Objective

Evaluate whether data-driven weather models (GraphCast/GenCast, Pangu, Aurora, FuXi, etc.) and AI nowcasting models can provide background fields for tornadoes/thunderstorms/typhoons, serve wind power forecasting, offer utility for CFD, and assess the value of reforecasts for our research. Deliver an assessment table, a literature review audit ID, a minimal experiment, and a one-sentence conclusion. Task brief approved by the PI on 10-13.

# Current status

10-13 14:00 **On hold** (feasibility questions fully answered). Three-event reforecast (Wister, Akron, ALTU): maximum lead time simultaneously satisfying all three pre-specified criteria is 12/12/none for Pangu and 24/none/12 h for FuXi, showing instability; the bottleneck is CAPE (on ALTU, Pangu fails MLCAPE across all five lead times, with errors 1.2–11× that of persistence), whereas wind shear and DCAPE outperform persistence across most combinations. Literature review audit verdict issued. Total compute across both experiment phases is roughly estimated at <11.3 CU (≤20 cap). Figures with source footnotes have been generated (previous figures retained).

**Reason for putting on hold**: Because we focus on historical event hindcasts, using ERA5/HRRR analysis fields directly for the environment is superior; when using AI global models directly as downburst background fields, the thermodynamic environment only holds up to 12–24 h and is unstable, while 0.25° cannot resolve the storm itself; the bottleneck in the CM1 track lies in observational candidates rather than environmental data. **What is missing / when to restart**: (1) Our group pivots toward forecasting tasks (assessing severe convective wind environments several days ahead, or wind power forecasting); (2) The WRF track is willing to test "Pangu-driven WRF vs. GFS-driven WRF" (precedents: Xu 2024 ERL, Liu 2024 JGR:MLC), which is the most promising direction but requires formal project scale; (3) When multi-event statistics are needed, start with existing AWS AIWP reforecasts (from 2022-01 onward, GFS/IFS initial conditions at 00/12 UTC, subject to verifying that pressure-level variables are included); (4) If CAPE is the target, switch to 37-level outputs or a post-processing model (such as the Transformer post-processor in Hua et al., arXiv 2505.11750). A restart would cost ~2–3 CU per event.

# Decisions made

- 2026-10-13 Literature review audit verdict: GraphCast weights open for commercial use starting 2026-08; NowcastNet–HRRR comparison updated to cite Das 2024; reforecast interpretation restricted to qualified scope
- 2026-10-13 Minimal experiment criteria and conventions (13-level ERA5 as primary reference, diurnally aligned persistence baseline, all three criteria satisfied simultaneously)
- 2026-10-13 Kickoff plan and minimal experiment direction (multi-initialization reforecasts rather than single-time comparisons)

# Assessment table (finalized after review verdict, 10-13; recommendation level reflects internal priority judgment)

| Question | Our specific use | Existing data / precedents | Minimal experiment | Cost | Recommendation |
|---|---|---|---|---|---|
| 1 Severe convective environmental background fields (directly using AI global models) | Assess favorable environments in advance and decide whether to schedule WRF/CM1 runs; not used for CM1 multi-event screening (CM1 lead) | Our group's reforecasts; Feldmann 2024 arXiv 2406.09474, Hua 2025 GRL 10.1029/2024GL109611 (abstract) | Completed on three events: under our n=3, domain-maximum, diurnally aligned persistence criteria, maximum usable lead time is Pangu 12/12/none, FuXi 24/none/12 h; wind shear and DCAPE beat persistence in most combinations, while MLCAPE/MUCAPE is the bottleneck | <11.3 CU | Low–Medium |
| 1′ AI large-scale fields driving WRF | Extend usable lead time of WRF driving fields | Xu 2024 ERL 10.1088/1748-9326/ad41f0 (Pangu→WRF heavy rain TS maintained for 8.5 days vs. ~3 days for GFS), Liu 2024 JGR:MLC 10.1029/2024JH000207 (typhoon intensity error reduced by 32% vs. global model and 23% vs. ERA5-driven WRF); CM1 can only ingest soundings | Untested, requires coordination with WRF track | — | Medium |

| Question | Our specific use | Existing data / precedents | Minimal experiment | Cost | Recommendation |
|---|---|---|---|---|---|
| 1b Convection-allowing AI and nowcasting | 0–12 h thunderstorms and gusts | StormCast arXiv 2408.10958, WoFSCast GRL 10.1029/2024GL112383 (numerical model emulators); FuXi-Nowcast arXiv 2512.08974 (includes gusts); NowcastNet third-party evaluation Das 2024 10.1038/s41612-024-00834-8 (CSI 0.30 vs. HRRR 0.04, >16 mm/h) | Not for now: no CAM training archive or radar training data | — | Medium (watch) |
| 1c Typhoons | Track / intensity background fields | DeMaria 2025 AIES 10.1175/AIES-D-24-0085.1 (track comparable to best operational models, intensity systematically biased weak); near-surface wind profiles require coupling with WRF or a similar reconstruction (unverified when coupled to AI forecasts) | As needed | — | Medium |
| 2 Wind power | Hub-height wind / power | GenCast Nature 10.1038/s41586-024-08252-9 (97.2%/1320; wind power evaluated via idealized power curve aggregation); Hardy 2025 10.1002/met.70038 (0.18 vs. 0.63 is contemporaneous ERA5 reconstruction; AIFS forecast outperforms IFS after ~Day 3) | Do not pursue (outside our group's main track) | — | Medium–High for wind power itself; Low for our group's main track |

| Question | Our specific use | Existing data / precedents | Minimal experiment | Cost | Recommendation |
|---|---|---|---|---|---|
| 3 CFD | Downscaling chain boundary conditions, inflow | FuXi-CFD Nat Commun 10.1038/s41467-026-70562-5 (30 m terrain wind including TKE, validated against met masts; trained on neutral steady RANS); urban microclimate CFM (relative to LES) | Do not pursue | — | Low–Medium (not applicable to transient downburst inflow) |
| 4 Reforecasts / predictability | Lead-time statistics, ensembles around historical events | AWS AIWP (Radford 2025 BAMS 10.1175/BAMS-D-24-0057.1; Pangu/FCNv2 from 2020-10, GraphCast operational from 2022-01, GFS/IFS initial conditions at 00/12 UTC); HENS GMD 18:5575; Vonich arXiv 2504.20238 (initial condition optimization given known future, not forecasting skill) | Candidate: fetch post-2022 multi-event data from AIWP (must first verify archive contains pressure-level variables) | Model-run-free, still requires data retrieval & diagnostics | Medium |

# What was tried and results

| Date | What was done | One-line result |
|---|---|---|
| 10-13 | Reforecasts with Pangu and FuXi (operational GraphCast training period overlaps the events and the ERA5 version lacks an off-the-shelf wrapper, so FuXi was used instead) for Wister 18 UTC and Akron 06 UTC at lead times of 12/24/48/72/120 h against 13-level ERA5, using diurnally aligned persistence as baseline and three pre-specified criteria | Maximum lead time satisfying all three criteria simultaneously: Wister Pangu 12 h, FuXi 24 h; Akron Pangu 12 h, FuXi none. Parameter by parameter: 0–6 km wind shear error at most lead times is only 0.1–0.6× persistence; CAPE at ≥24–48 h is **worse than persistence** (error ratio 1.5–4). Following the literal wording of the task brief, Criterion 2 averaged across mixed units of J/kg, m/s, and K, so it was effectively dominated by CAPE (a flaw in the task brief's wording, judged strictly as written without post-hoc modification). MLCAPE computed from 13 levels is ~500 J/kg lower than from 37 levels, making vertical resolution itself a major error source. FuXi mirror weights were not verified file-by-file against the original weights |
| 10-13 | Added third event ALTU 2023-07-03 18 UTC (supplied by the CM1 track) under the same criteria; also produced a three-event station grid point + 3×3 parameter table | Pangu failed across all five lead times (rejected on MLCAPE alone at every lead time), while FuXi passed only at 12 h; 13-level MLCAPE is 532 vs. 37-level 797 J/kg; model "surface dewpoint" is a proxy taken from the lowest pressure level |
| 10-14 | Sample expansion: AWS AIWP reforecasts (Pangu, GraphCast Operational), 40 ASOS ≥50 kt thunderstorm high-wind days in 2023–24, parameter-by-parameter criteria | DCAPE and 0–1 km wind shear passed across 12–120 h; 0–6 km wind shear passed across all lead times for Pangu and up to 72 h for GraphCast; MUCAPE and MLCAPE failed across all lead times (event-by-event wins 14–25/40) |

- 10-21 progress and plan report issued.

# Open items and next steps

- 10-13 Intra-track trial rule (10-13 to 10-20): Upon restart, batch pure-QA routine tasks with fixed question banks (abstract initial screening, single-value extraction from papers, table field extraction, initial tag classification) are delegated by default, with 10% re-checked by Claude to record the agreement rate; task types with agreement <90% revert to the original workflow; does not apply to reforecasts, diagnostics, scientific judgment, or review submissions.

- When the literature review (default tier) returns: verify citations, then finalize the assessment table after the verdict.
- The ideas agent has returned (10-13 12:30), and the lead has re-verified the parameter-by-parameter error ratios. Finalize Row 1 once the ALTU supplemental run returns.
- CM1 has replied (10-13): not currently working on this; offers limited help for their multi-event validation and serves only as environmental context; provided ALTU 2023-07-03 and the station parameter list → added into the experiment (queued after the earlier run, criteria unchanged).
- Event definition sources: WRF lead reply on 10-13; Akron based on the source PDF pp. 1–2, 9–11.
- Experimental design note: Parent project review previously canceled the "single-event same-time AI field vs. ERA5" comparison (since AI models are trained on ERA5, that yields no new information). This experiment instead evaluates the growth of environmental parameter errors across **different lead times**, using persistence (ERA5 at initialization time) as the baseline; additionally, event dates must be checked to ensure they fall outside each model's training period (GraphCast/Pangu training ends in 2017, placing Wister 2018 in the test period, pending verification against the original papers).

## Literature basis

- 10-13 showcase page citations have been added to the "Forecasting uses of AI weather models" section (ERA5/ARCO, Pangu, FuXi, GraphCast, earth2studio, ai-models, MetPy, plus the 18 papers underlying the assessment table); reforecast directory added, and new figures with source footnotes have been generated.
- Parent project page "Ancillary outputs" row 2b: Feldmann 2024 (arXiv 2406.09474), Hua 2025 GRL (10.1029/2024GL109611), not yet added to the literature library as cards.
- 220 m global LES is a **physics-based model**, not AI; abstract-only verification.
- Generative 3D instantaneous urban wind and temperature fields, abstract only.
- Physics-constrained network for typhoon near-surface wind profiles, abstract only.
- AI forecasting + proprietary observation network, superiority claims unverified.
- Weather model repurposed as a forced climate emulator (arXiv 2605.29976).
- Open source (row 2b): earth2studio, ai-models under Apache-2.0; licensing terms for Pangu and Aurora weights pending verification.

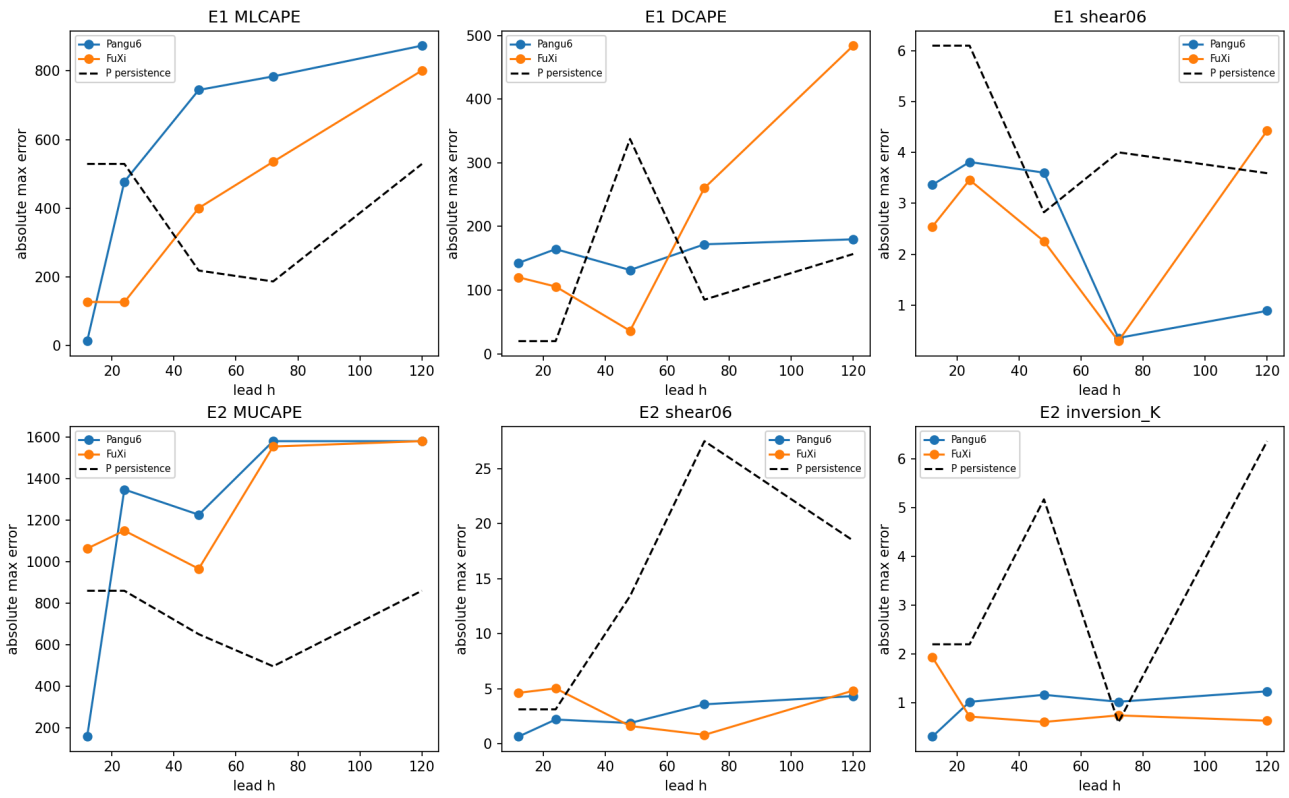
## PI notes

- 10-13 (key points from the PI's own words): Useful for forecasting (especially nowcasting/short-range), capable of providing background fields for tornadoes, thunderstorms, and typhoons; should be even more useful for wind power; asked whether such foundation models also hold reference value for CFD. Approved opening: "Go ahead and open it, you drive it forward."

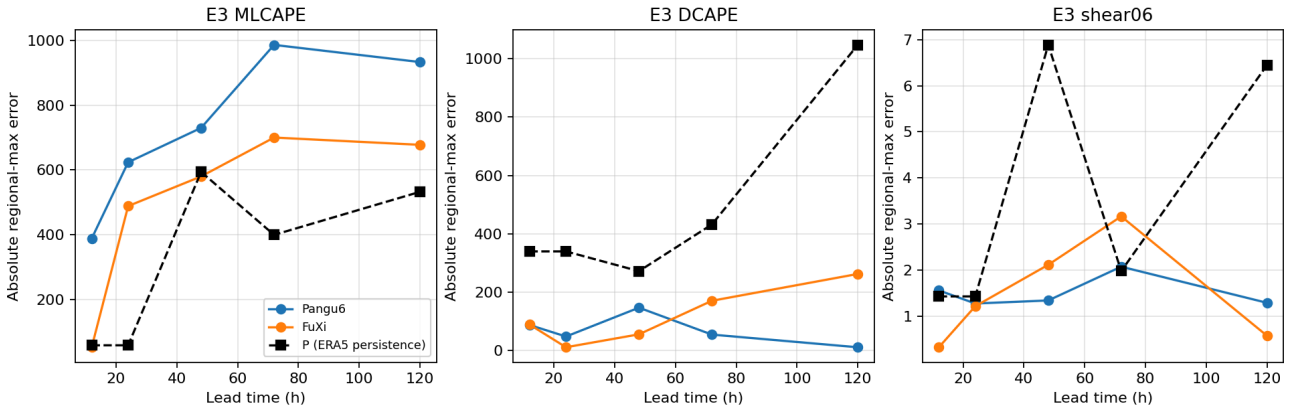
# 10-22 seventh verification batch

- **Conclusion scope must be narrowed**: "AI global models provide convective background fields only out to 12–24 h" applies only to the three events in this project and when evaluated by local environmental parameters (such as CAPE). White 2026 (AIES, 10.1175/aies-d-25-0065.1) uses Pangu environmental fields + random forests to forecast Days 1–8 severe convective hazard probabilities (wind/hail/tornado) tested on 2024, achieving BSS>0 at all lead times and remaining comparable to GEFS-MLP out to Day 7 (published PDF p.7, spot-checked and confirmed by the lead); Hua 2026 reports similar results (Days 3–8). Both forecast regional hazard occurrence probabilities rather than local environmental parameters or station gusts, so they are not same-metric counterexamples, but they show that the "AI background field + statistical post-processing" pathway has skill at medium-range lead times.
- All other judgments in this project (StormCast configuration below threshold, v2 scoped gap) remain unchanged; ProbSevere v3, Jahan 2024 (WRF + ML gusts), and Hallgren 2024 (lidar-trained local hub-height wind) serve as adjacent precedents.

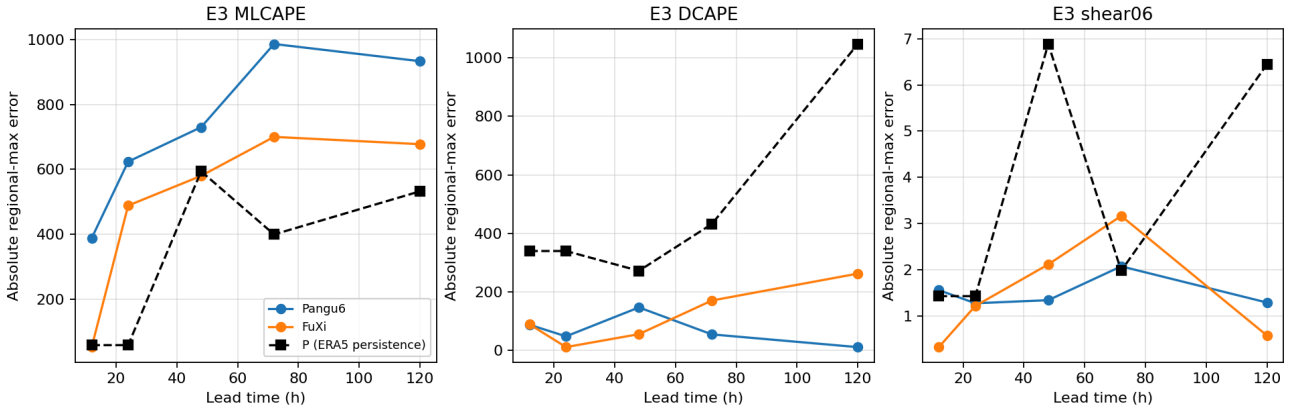
# Key figures




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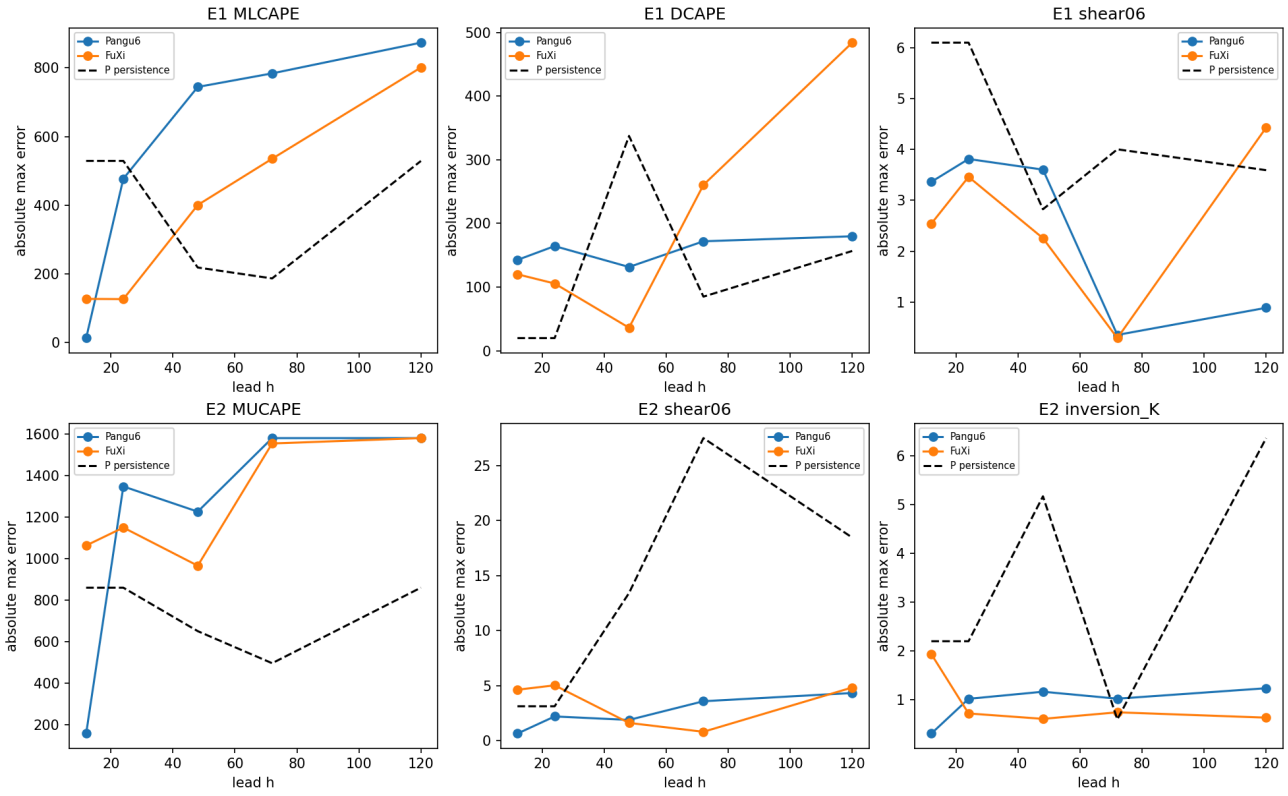



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数据：ERA5（Hersbach et al. 2020, C3S/ARCO）；模型：Pangu-Weather、FuXi（earth2studio）

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数据：ERA5（Hersbach et al. 2020, C3S/ARCO）；模型：Pangu-Weather、FuXi（earth2studio）

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